NetDiagnose AI

Explainable Home and Last-Mile Connectivity Diagnosis

Hackathon: AIORI-3 | **Problem Statement:** PS-015 | **Team Code:** A3-PS015-TC437
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Project Overview

NetDiagnose AI is a proposed web-based system designed to help home internet users understand and investigate common connectivity problems.

Internet users may experience slow browsing, unstable Wi-Fi, high latency and frequent disconnections. However, identifying the possible cause can be difficult without technical knowledge or access to appropriate diagnostic information.

The proposed system will combine user-reported symptoms with available network measurements to identify possible connectivity issues. It will provide understandable explanations, supporting evidence and practical troubleshooting recommendations.

The project aims to make basic network diagnosis more accessible to non-technical users through a simple and explainable interface.

Project Objective

To develop a user-friendly connectivity diagnosis tool that helps users understand possible internet problems and take appropriate troubleshooting actions.

1. Introduction

Internet connectivity has become an important part of everyday life. Students depend on the internet for online classes, employees use it for remote work, and households rely on it for communication, entertainment and digital services.

Despite its importance, home internet connectivity is not always stable. Users may experience slow speeds, interruptions, inconsistent performance and difficulty accessing online services.

Identifying the source of these problems can be challenging. A connectivity issue may be related to the user's device, wireless signal, router, local network conditions or internet service provider.

Non-technical users may not understand measurements such as latency, packet loss and upload or download speed. As a result, they may struggle to determine which troubleshooting steps are appropriate.

NetDiagnose AI proposes a structured approach to investigating these problems.

The system will collect information about the user's symptoms and available network measurements. It will analyse this information using predefined diagnostic rules and present possible causes with explanations.

Rather than presenting a diagnosis as an unquestionable fact, the system will communicate uncertainty when the available evidence is insufficient.

Project Vision

To make home internet troubleshooting simpler, more transparent and easier to understand.

Project Mission

To connect network symptoms and available measurements with understandable explanations and actionable troubleshooting recommendations.

2. Problem Statement

Home internet users frequently encounter connectivity problems but may lack a clear and accessible way to investigate their causes.

Existing network measurements can provide useful information, but interpreting these measurements often requires technical knowledge. Users may not know whether a problem is related to Wi-Fi, their router, their device or their internet service.

This creates a gap between detecting a connectivity problem and understanding what to do next.

NetDiagnose AI proposes to address this gap by combining reported symptoms and available measurements with transparent diagnostic rules.

2.1 Major Challenges

1. Limited Diagnostic Understanding

Users may recognise that their internet is slow but may not understand the possible reasons.

2. Difficulty Interpreting Network Measurements

Measurements such as latency and packet loss can be difficult to interpret without technical guidance.

3. Unstructured Troubleshooting

Users may try multiple troubleshooting actions without following a clear sequence.

4. Uncertainty About the Source

Similar symptoms can arise from different causes. For example, slow internet may result from wireless interference, congestion or an issue beyond the local network.

5. Communication With ISP Support

Users may not have organised information about their symptoms and measurements when contacting technical support.

2.2 Need for the Proposed System

A system that presents possible causes, supporting evidence and recommended actions could help users investigate connectivity problems more systematically.

3. Existing Approaches and Their Limitations

Users commonly rely on several methods to investigate internet connectivity problems.

3.1 Manual Troubleshooting

Users may restart their router, reconnect their devices or move closer to the Wi-Fi access point.

Limitation: These actions may help, but they do not always explain the underlying cause.

3.2 Internet Speed Testing

Speed-testing tools measure aspects of connection performance, including download speed, upload speed and sometimes latency.

Limitation: A speed test provides performance measurements but does not necessarily identify the underlying cause of poor performance.

3.3 Router Status Information

Routers may provide information about connected devices, signal conditions and network status.

Limitation: The availability and clarity of this information vary between router models.

3.4 ISP Customer Support

Users can contact their internet service provider for assistance.

Limitation: Users may find it difficult to describe the issue precisely or provide useful measurements.

3.5 Proposed Improvement

NetDiagnose AI aims to complement these existing approaches by bringing user-reported symptoms, available measurements, possible explanations and troubleshooting guidance into a single workflow.

The proposed system will not replace professional network diagnostics or ISP support. It will help users organise the information available to them.

4. Proposed Solution: NetDiagnose AI

NetDiagnose AI is a proposed web-based connectivity diagnosis system.

The system will guide users through a structured assessment of their internet problems and present possible explanations based on available evidence.

4.1 Core Components

1. User Input Module

Collects details about the user's connectivity symptoms, such as:

- Slow internet speed
- Frequent disconnections
- Unstable Wi-Fi
- Difficulty accessing websites
- Delays during video calls
- Poor performance on specific devices

2. Network Measurement Module

Uses measurements supplied by the user or obtained through supported testing methods. Possible measurements include download speed, upload speed, latency, packet loss and wireless signal information where available.

3. Diagnostic Rule Engine

Evaluates the collected information against predefined rules to identify possible causes.

4. Explanation Module

Displays the symptoms and measurements that support each possible diagnosis.

5. Recommendation Module

Provides troubleshooting steps relevant to the identified symptoms.

6. Results Interface

Presents the assessment in a simple format that non-technical users can understand.

4.2 Expected Result

The user receives a structured diagnostic summary containing the reported problem, available evidence, possible causes, recommended actions and any limitations of the assessment.

5. Project Objectives and Scope

The primary objective is to design and develop an accessible tool for investigating common home internet connectivity problems.

5.1 Specific Objectives

Objective 1: Symptom Collection

Provide a simple interface for recording connectivity problems.

Objective 2: Measurement-Based Assessment

Use available network measurements to support the diagnostic process.

Objective 3: Explainable Diagnosis

Present possible causes with explanations rather than displaying only a generic error.

Objective 4: Guided Troubleshooting

Recommend relevant actions in a logical sequence.

Objective 5: Uncertainty Handling

Indicate when available information is insufficient to determine a likely cause.

Objective 6: Accessible Interface

Present technical information using clear and understandable language.

5.2 Project Scope

The initial version will focus on common home internet and last-mile connectivity symptoms. The planned scope includes:

- Symptom reporting
- Entry or collection of supported measurements
- Rule-based assessment
- Explainable results
- Troubleshooting recommendations
- Guidance for contacting the ISP

5.3 Scope Limitations

The initial system will not be assumed to have direct access to every router, ISP network or external network device. It will not guarantee that every connectivity problem can be identified accurately. The reliability of its results will depend on the quality and availability of the input information and the diagnostic rules.

6. Proposed System Architecture

The proposed system will consist of a frontend, backend, diagnostic rule engine and results interface.

6.1 Frontend Layer

The frontend will provide the interface through which users enter symptoms and view diagnostic results. **Proposed technologies:** HTML, CSS and JavaScript.

6.2 Backend Layer

The backend will receive user inputs, process requests and coordinate the diagnostic workflow. **Proposed technology:** Python with Flask.

6.3 Diagnostic Engine

The diagnostic engine will compare the collected symptoms and measurements against predefined rules. It will generate possible causes only when the available information supports those conclusions.

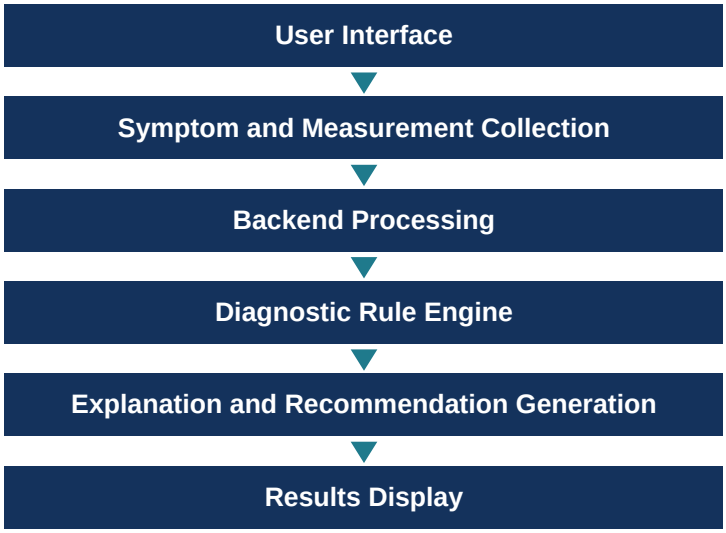
6.4 Explanation Layer

This component will present the evidence behind the assessment. For example, if the available measurements show high latency, the system may explain that delayed network responses could affect interactive activities. High latency alone, however, would not establish the exact source of the problem.

6.5 Recommendation Layer

The recommendation component will map identified symptoms and possible causes to appropriate troubleshooting actions.

6.6 System Flow



6.7 Design Principle

The architecture will prioritise simplicity, explainability, modularity and the ability to improve diagnostic rules as the project develops.

7. Proposed Technology Stack

The following technologies are proposed for the initial implementation.

7.1 HTML

HTML will define the structure of the web pages, including input forms, buttons and result sections.

7.2 CSS

CSS will be used to style the interface and improve readability, spacing, responsiveness and visual consistency.

7.3 JavaScript

JavaScript will support interactive behaviour, input validation and communication between the frontend and backend.

7.4 Python

Python will implement the backend logic and process the information submitted by users.

7.5 Flask

Flask will provide the web application framework for handling requests and returning diagnostic results.

7.6 Rule-Based Diagnosis

The initial diagnostic engine will use predefined conditions rather than assuming that a trained AI model already exists. These rules will map selected symptoms and measurements to possible causes and appropriate recommendations.

7.7 GitHub

GitHub will support version control, source-code management and collaboration between team members.

7.8 Why This Technology Stack?

The proposed stack is suitable for a small web-based prototype because it uses commonly available development technologies and allows the interface, backend and diagnostic logic to be developed separately.

7.9 Future Enhancements

Future versions may incorporate more advanced network testing, historical performance analysis or machine-learning methods if suitable data, validation and implementation resources become available.

> **Note:** These are proposed technologies. The final technology stack should reflect the tools actually implemented in the project.

8. Proposed Diagnostic Methodology

NetDiagnose AI will follow a structured workflow to investigate connectivity problems.

Step 1: Identify the Symptom

The user selects the problem they are experiencing. Examples include slow internet, frequent disconnections or delays during video calls.

Step 2: Collect Available Information

The system collects the selected symptoms and any available measurements. These may include download speed, upload speed, latency and packet loss.

Step 3: Evaluate the Evidence

The diagnostic engine compares the available information against predefined rules. It considers whether the observed measurements and symptoms are consistent with particular connectivity issues.

Step 4: Identify Possible Causes

The system generates possible explanations based on the evidence. For example, weak wireless signal information may support investigating Wi-Fi coverage. High latency may indicate delayed network responses, but further testing may be required to identify the cause.

Step 5: Explain the Assessment

The system presents the relevant evidence and explains why a particular cause is being considered.

Step 6: Recommend Troubleshooting Actions

The system suggests relevant next steps, such as checking wireless signal conditions, testing another device or repeating a network measurement.

Step 7: Escalate When Necessary

If the issue remains unresolved or appears to require external investigation, the system recommends contacting the ISP.

Methodology Summary

Identify > Collect > Analyse > Explain > Recommend > Reassess

This approach is intended to make the troubleshooting process more systematic and understandable.

9. Innovation, Benefits and Expected Outcomes

9. Innovation and Uniqueness

The proposed system focuses on connecting network symptoms with understandable explanations and practical actions.

9.1 Explainability. The system aims to show which reported symptoms or measurements support a possible diagnosis.

9.2 Evidence-Based Recommendations. Recommendations will be linked to the available information instead of being presented as unrelated general advice.

9.3 User-Friendly Communication. Technical results will be explained in language that is accessible to users without networking expertise.

9.4 Uncertainty Awareness. When the available evidence cannot distinguish between multiple causes, the system will communicate that limitation and recommend further testing.

9.5 Structured Troubleshooting. The workflow will organise the investigation into clear steps rather than presenting an unstructured list of possible fixes.

10. Expected Benefits

For Home Users: Improved understanding of common connectivity symptoms and possible next steps.

For Students and Remote Workers: A more structured way to investigate connectivity problems that affect online activities.

For Non-Technical Users: Simpler explanations of network measurements and potential issues.

For ISP Communication: Better organisation of symptoms and available measurements when requesting support.

For Future Development: A modular foundation that can be expanded with additional measurements and diagnostic methods.

10.1 Expected Outcomes

The expected outcome is a functional prototype capable of collecting selected inputs, evaluating predefined diagnostic rules and displaying explanations and recommendations. Actual performance, accuracy and usability will need to be assessed through testing.

> No numerical improvement or diagnostic accuracy should be claimed before it has been measured.

11. Testing, Validation and Limitations

Testing will be necessary to determine whether the system behaves as intended.

11.1 Functional Testing. Verify that users can enter symptoms, submit information and receive results.

11.2 Rule Testing. Check whether each diagnostic rule produces the expected output for defined input conditions.

11.3 Explanation Testing. Verify that the explanations accurately reflect the evidence used by the diagnostic engine.

11.4 Recommendation Testing. Check whether the recommended troubleshooting actions are relevant to the symptoms and possible causes identified.

11.5 Usability Testing. Ask users to complete basic diagnostic tasks and provide feedback about the clarity of the interface and recommendations.

11.6 Uncertainty Testing. Provide incomplete or conflicting inputs and verify that the system communicates uncertainty instead of producing an unsupported conclusion.

11.7 Example Test Scenarios

Scenario 1: Slow Internet. *Input:* The user reports slow internet performance. *Expected:* The system requests or evaluates available measurements and suggests relevant checks without assuming a single cause.

Scenario 2: High Latency. *Input:* Available testing indicates high latency. *Expected:* The system explains the potential effect of delayed responses and recommends further investigation.

Scenario 3: Frequent Wi-Fi Disconnections. *Input:* The user reports repeated wireless disconnections. *Expected:* The system suggests investigating wireless conditions, device behaviour and router status where applicable.

Scenario 4: Insufficient Information. *Input:* The user reports a problem but provides insufficient measurements. *Expected:* The system identifies the limitation and requests additional information or testing.

11.8 Limitations

- Not every connectivity problem can be diagnosed from user inputs alone.
- Available measurements may vary by device and testing method.
- Similar symptoms may have different underlying causes.
- The initial rule-based system may not cover every network condition.
- Recommendations require validation before the system can be relied upon in real-world situations.

12. Future Scope

The project may be extended in several directions after the initial prototype is implemented and evaluated.

12.1 Automated Network Testing. Future versions could integrate supported tools to collect network measurements automatically.

12.2 Historical Performance Analysis. The system could maintain user-authorised measurement history to help identify recurring patterns.

12.3 Advanced Diagnostic Methods. Additional diagnostic rules or machine-learning methods could be explored if suitable datasets and validation procedures are available.

12.4 Expanded Device Support. Future development could explore compatibility with additional operating systems, routers and network diagnostic interfaces.

12.5 Improved User Guidance. The system could provide more contextual recommendations based on the device, connection type and available measurements.

13. Conclusion

NetDiagnose AI proposes an explainable and user-friendly approach to investigating home internet connectivity problems. The project aims to bridge the gap between observing a connectivity issue and understanding the next steps by combining reported symptoms, available measurements, transparent diagnostic rules and practical recommendations.

Its initial implementation will focus on a simple web-based prototype using commonly available technologies. The reliability and usefulness of the proposed system will be evaluated through functional testing, rule validation and user feedback.

The long-term goal is to make basic connectivity diagnosis more accessible while clearly communicating uncertainty and recognising when professional assistance is required.

14. References

1. **AIORI-3 Hackathon Guidelines** - Official hackathon instructions and submission requirements. https://portal.aiori.in/aiori-3/hackathon-guideline/

2. **AIORI-3 Selected Teams** - Official selected-team information. https://portal.aiori.in/aiori-3/selected-teams/

3. **Internet Engineering Task Force (IETF)** - Technical standards and documentation for Internet protocols. https://www.ietf.org/

Team Information

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